

Guideline Title: Malignant Hyperthermia

Guideline Number: GUID-3203-M020

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
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- I. **PURPOSE:** Recognize malignant hyperthermia, a life-threatening disorder of the skeletal muscle, which can be triggered by inhalational anesthetics and Succinylcholine.
- II. **DEFINITIONS:**
- III. **DEPARTMENT GUIDELINE:**
 - a. Establish care as defined by *GUID3203-G002 – General Patient Care*.
 - b. Assess for hypoxemia, tachycardia/cardiac dysrhythmias, muscle rigidity or failure of the masseter muscles to relax for intubation, hot dry skin/hyperthermia, hypotension, sustained tetanic muscle rigidity/contractions, hypermetabolism, muscle cell destruction, hyperkalemia, tachypnea from severe respiratory and metabolic acidosis, & cyanotic mottling of the skin.
 - c. Stop all paralytics/anesthetic agents.
 - d. Hyperventilate with 100% oxygen to flush volatile anesthetics and lower ETCO₂.
 - e. Initiate immediate cooling.
 - f. Infuse 0.9% NS IV with a urinary output goal of >1 ml/kg/hr.
 - g. If at a sending facility, consult with physician and request **Dantrolene 2.5mg/kg IV**.
 - i. Administer rapidly through a large-bore IV if possible.
 - ii. Dantrium/Revonto brand: Each 20 mg vial of should be reconstituted by adding 60 ml of sterile water for injection and the vial shaken until the solution is clear (20 seconds).
 1. Reminder: this drug contains 3 grams of mannitol in each vial
 - iii. Ryanodex brand: Each 250 mg vial should be reconstituted with 5 ml of sterile water and shaken to ensure orange-colored uniform, opaque suspension.
 - iv. Repeat as frequently & as needed until patient responds with a decrease in ETCO₂, muscle rigidity, and/or heart rate. Large doses (> 10 mg/kg) may be required if persistent rigidity. If giving large doses (> 10 mg/kg) without symptom resolution, consider alternative diagnoses.
 - h. If actively transporting a patient, divert to the nearest facility with capability of administering immediate dose of Dantrolene.
- IV. **DOCUMENTATION:** EMS Charts
- V. **REFERENCES:**
 - a. Malignant Hyperthermia Association of the United States. (2020). www.mhaus.org